MTH309 PRACTICE PROBLEMS WEEK 9

Question 1. Let $\{A_n\}_n$ be a sequence of sets in a measure space $(\Omega, \mathcal{F}, \mu)$ such that $\liminf_n A_n = \limsup_n A_n =: A$. Is it true that $\lim_n \mu(A_n) = \mu(A)$?

Question 2. Prove the following version of Markov inequality. Let $f : \Omega \to \bar{\mathbb{R}}$ be Borel measurable. Then for any $A \in \mathcal{F}$ and $c > 0$ show that

$$\mu(\{|f| \geq c\} \cap A) \leq \frac{1}{c} \int_A |f| \, d\mu.$$

Question 3. Fix $1 \leq p < q < \infty$. Show that

(i) Show that $\mathcal{L}^q(\mathbb{P}) \subset \mathcal{L}^p(\mathbb{P})$ with $\|f\|_p \leq \|f\|_q, \forall f \in \mathcal{L}^q(\mathbb{P})$.

(ii) $X_n \xrightarrow[n\to\infty]{\mathcal{L}^q(\mathbb{P})} X$ implies $X_n \xrightarrow[n\to\infty]{\mathcal{L}^p(\mathbb{P})} X$.

Question 4. Fix $1 \leq p < q < \infty$. Given any finite measure μ on a measurable space $(\Omega, \mathcal{F})$, show that $\mathcal{L}^q(\mu) \subset \mathcal{L}^p(\mu)$.

Question 5. Fix $1 \leq p < q < \infty$. Let λ denote the Lebesgue measure on $(\mathbb{R}, \mathbb{B}_{\mathbb{R}})$. Find $f \in \mathcal{L}^p(\lambda), g \in \mathcal{L}^q(\lambda)$ such that $f \notin \mathcal{L}^q(\lambda), g \notin \mathcal{L}^p(\lambda)$.

Question 6. In each of the following cases, find examples of random variables $X, X_1, X_2, \cdots$ on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that the indicated property holds.

(i) $X_n \xrightarrow[n\to\infty]{\mathcal{L}^1(\mathbb{P})} X$, but $X_n \nrightarrow X$ a.s.

(ii) $X_n \xrightarrow[n\to\infty]{\text{a.s.}} X$, but $X_n \nrightarrow X$ in $\mathcal{L}^1(\mathbb{P})$.

(iii) Fix $1 \leq p < q < \infty$. $X_n \xrightarrow[n\to\infty]{\mathcal{L}^p(\mathbb{P})} X$, but $X_n \nrightarrow X$ in $\mathcal{L}^q(\mathbb{P})$.